

# Wi-Fi



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# Outline

WiFi

WiFi Security: WEP, WAP, WAP2

# Wi-Fi/WLAN

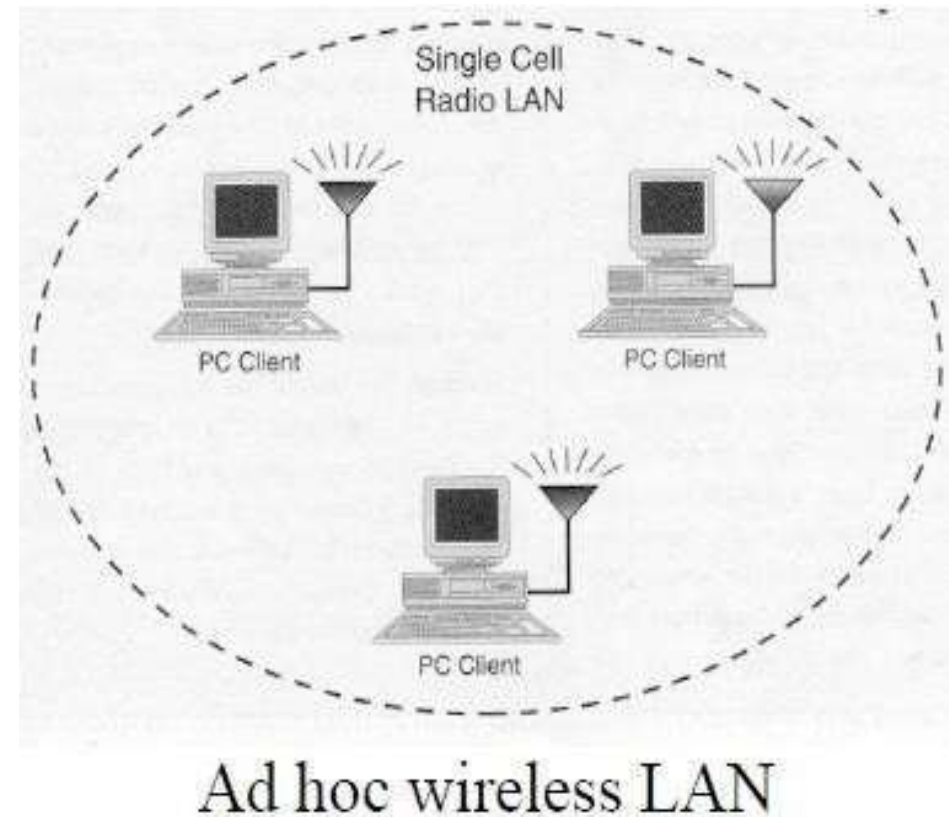
- Wireless Fidelity (Wi-Fi) allows us to connect to the Internet from anywhere (without wires) within the range of a base station.
- Types of Wi-Fi:
  - IEEE 802.11b : Max. transfer speed is 11Mbps, operate within 100-150 feet. Used in cordless phones and in microwave oven.
  - IEEE 802.11a: Max transfer speed is 54 Mbps, operates in 5GHz band, perform better in high density area.
  - IEEE 802.11g: operates within 100-150 feet, 2.4 GHz radio frequency, extremely fast, speed of 15-20 Mbps

# Introduction: WLAN

- A wireless LAN (WLAN) is a :
  - flexible data communication system implemented as an alternative for a wired LAN within a building or campus.
  - combine data connectivity with user mobility without losing connectivity.
- Flexible:
  - reconfigure /add more nodes to the network without much planning effort and cost of re-cabling.
- Two types of network topologies:
  - ad hoc and infrastructure.

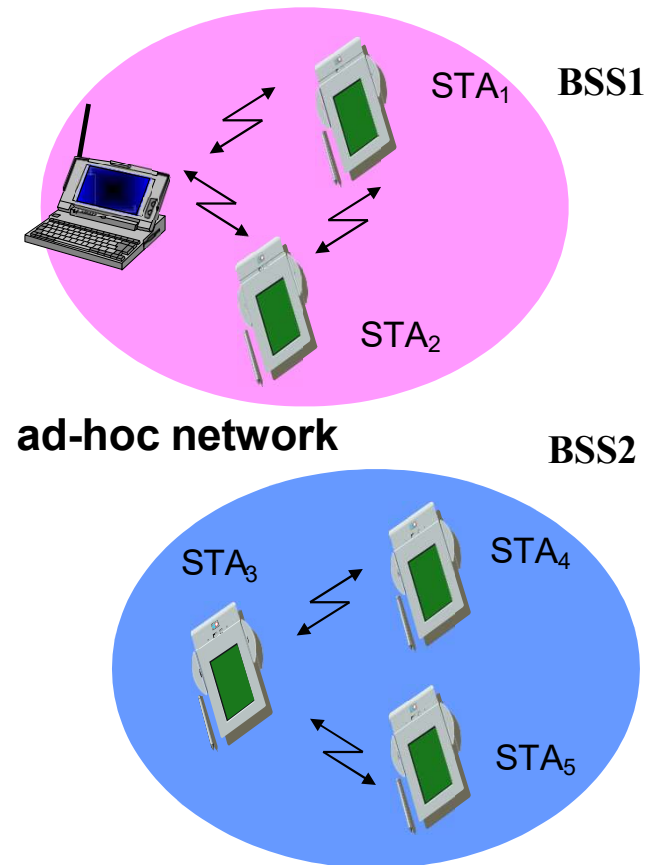
# WLAN Topology : Ad hoc

- An adhoc topology supports p2p connectivity where mobile nodes communicate directly with each other using wireless adapters.
- Ideal for meetings or setting up of temporary workgroups.
- It has disadvantage of limited coverage area



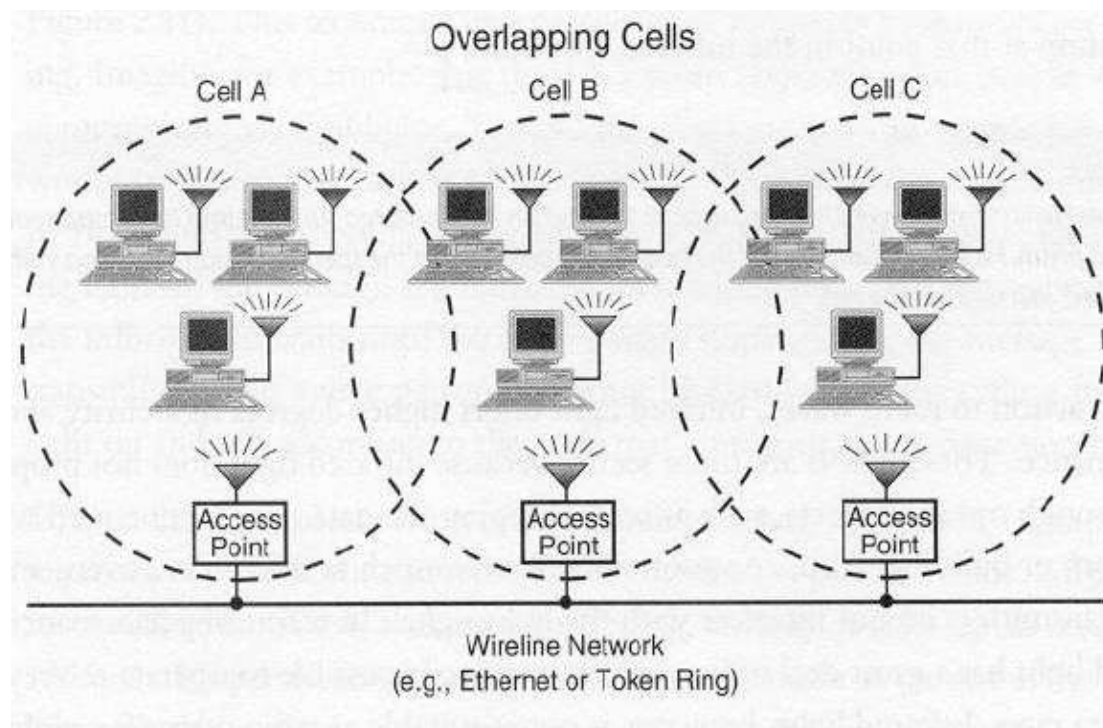
# Ad hoc network architecture

- The **basic service set (BSS)** consists of group of stations using the same radio frequency and is the basic building block of an IEEE 802.11 LAN.
- The **Independent BSS (IBSS)** is the simplest LAN. It may consist of as few as two stations.
- **IBSS** is also called the ad hoc mode in WLAN.
- The ovals can be thought of as the coverage area within which member stations can directly communicate.
- **Station (STA)** is the terminal with access mechanisms to the wireless medium and can directly communicate with a limited range.



# WLAN Topology : Infrastructure

- Use an access point that controls :
  - transmission and assigns priority levels
  - monitors network load and
  - allows mobile stations to roam from cell to cell.
- It handle traffic from a mobile station to another mobile station within the coverage area or to the wired backbone.

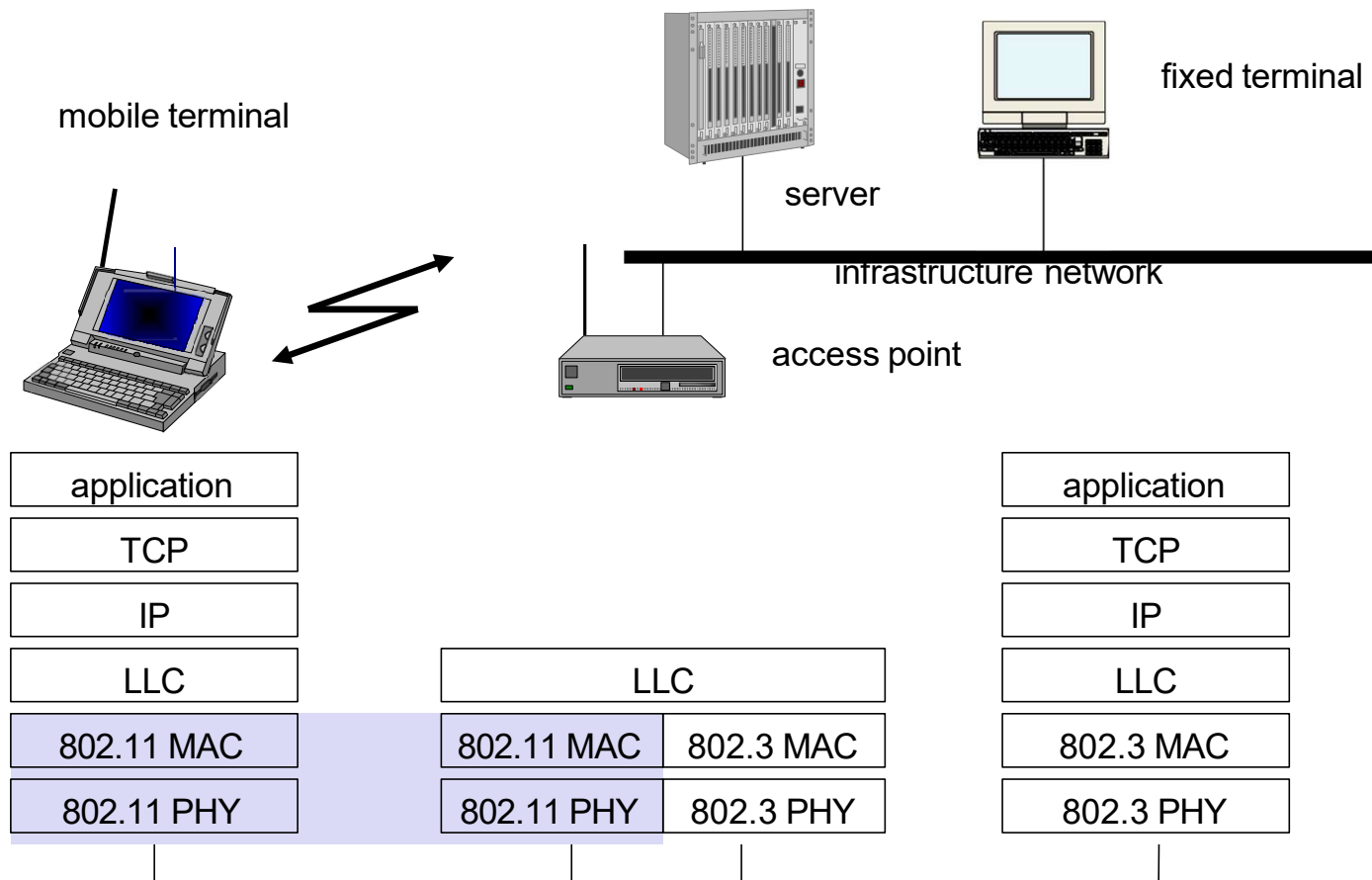


# Infrastructure WLAN components

- **Station (STA)**: terminal with access mechanisms to the wireless medium and radio contact to the access point
- **Basic Service Set (BSS)**: group of stations using the same radio frequency
- **Access Point**: station integrated into the wireless LAN and the distribution system
- **Portal**: bridge to other (wired) networks
- **Distribution System**: interconnection network to form one logical network (ESS: Extended Service Set) based on several BSS



# WLAN in the TCP/IP stack



# Wi-Fi: Security

## Types of Wireless Security

- Unsecured
- WEP
- WPA
- WPA2

## Why use wireless encryption?

## Additional Security Measures for your router

# Unsecured

- A wireless network with no sort of encryption algorithm applied.
- Any user can readily authenticate and access the internet.
- Packets are unencrypted and visible.
- **Attacks:**
  - **ARP Spoofing** - Associate attacker's MAC address with default gateway's IP. All traffic meant for gateway goes through attacker's machine first. Traffic can be passed through (passive sniff) or modified and passed (MIM).
  - **Firesheep** - Firefox extension that decodes cookies on unsecured network. Allows log in as user for sites like Facebook and Twitter.

# WEP: Wired Equivalent Privacy

- Deprecated security algorithm for IEEE 802.11 networking.
- Introduced as part of original 802.11 protocol in 1997.
- Standard 64 bit - WEP uses 40 bit key. Other 24 bits is IV.
- Can also use 128/256 bit protocols.
- **IV (Initialization Vector)** - prepended onto packets and is based on pre-shared key.
- Such short IVs in 64 bit caused reuse of IVs with same key, which significantly shortened key cracking times of WEP.
- **Attacks:**
  - **Aircrack-ng** - Linux command line tool. Sniffs packets on a network to obtain IVs and breaks WEP key using information present in the IVs. Can be done in less than 10 minutes.

# WPA: Wi-Fi Protected Access

- Flaws in WEP known since January 2001 - flaws include weak encryption (keys no longer than 40 bits), static encryption keys, lack of key distribution method.
- In April 2003, the Wi-Fi Alliance introduced an interoperable security protocol known as WiFi Protected Access (WPA).
- WPA was designed to be a replacement for WEP networks without requiring hardware replacements.
- WPA provides stronger data encryption (weak in WEP) and user authentication (largely missing in WEP).
- Also supported **AES encryption**
- Designed to be backward compatible with older hardware after firmware upgrades
- 4-Way Handshake and Group Key Handshake

# WPA: Wi-Fi Protected Access II

- WPA includes Temporal Key Integrity Protocol (TKIP) and 802.1x mechanisms.
- The combination of these two mechanisms provides dynamic key encryption and mutual authentication
- TKIP adds the following strengths to WEP:
  - Per-packet key construction and distribution: WPA automatically generates a new unique encryption key periodically for each client. This avoids the same key staying in use for weeks or months as they do with WEP.
  - Message integrity code: guard against forgery attacks.
  - 48-bit initialization vectors, use one-way hash function instead of XOR
- **"Beck-Tews Attack" - TKIP Exploit:**
  - PhD Candidate in Germany discovered a method for injecting small packets into a network using WPA and TKIP
  - Does not reveal full network key though, but can be used to spoof ARP and DNS packets

# WPA2: Wi-Fi Protected Access 2

- In July 2004, the IEEE approved the full IEEE 802.11i specification by WiFi Alliance known as WPA2.
- Released by Wi-Fi Alliance as upgrade to WPA and Backward compatible with WPA
- Required support of TKIP and AES instead of RC4
- Strong encryption and authentication for infrastructure and ad-hoc networks (WPA1 is limited to infrastructure networks)
- WPA2 certification has become mandatory for all new equipment certified by the Wi-Fi Alliance, ensuring that any reasonably modern hardware will support both WPA1 and WPA2.
- **"Hole 196" Attack:**
  - Allows already authenticated user to spoof mac address of router using the Group Temporal Key (known to all clients)
  - Client responds using their Pairwise Transient Key, which is unique to them, allowing attacker to decrypt the clients' packets

# Why does security matter?

- Unencrypted networks or exploitable encryption schemes allow hackers to:
  - Steal login credentials
  - Hijack browser sessions by stealing session cookies
  - Spoof packets on your network
- Use your network for malicious activity (ie Spam, DDOS)
  - Authorities will charge you with the crimes because it's your network



# Other Security Measures

- Enable MAC Address filtering
  - Prevents unauthorized computers from gaining access even if they have the correct network key
- Enable router firewall
- Change default Network SSID to something obscure
- Change default router password
- Change encryption password frequently

# References

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